i7Lab 4: Secure Connect and Communication Platform

Course: Web Dev III

Unit Covered: Unit – 4

Marks: 2.5

Mode: 100% In-Class (Lab Practicals)

Time: 30–40 minutes

Assignment Overview

The Secure Connect and Communication Platform is a hands-on lab designed to strengthen your understanding of backend security and real-time communication using Node.js. In this lab, you will build a secure server-side application that allows users to sign up, log in, access protected resources, manage personal notes, and communicate in real time without relying on any pre-built authentication or chat frameworks.

This assignment focuses on real-world backend operations, authentication flows, and event-driven communication techniques that are essential for full-stack development.

Learning Objectives

- Understand and apply core authentication and authorization concepts
- Securely hash and verify passwords using bcrypt
- Generate, verify, and manage JWT tokens for stateless sessions
- Implement protected routes and role-based access control (RBAC)
- Build and secure a Notes API with proper CRUD operations
- Handle real-time, bidirectional communication using WebSockets / Socket.IO
- Implement event-driven features using Node.js EventEmitter
- Manage environment variables securely using dotenv

Technology Stack

- Node.js – Server-side runtime environment
- Express.js – Backend routing and middleware
- JWT (jsonwebtoken) – Authentication tokens
- bcrypt – Password hashing
- dotenv – Environment variable management
- Socket.IO – Real-time, event-driven communication
- MongoDB / any preferred database – Data persistence

Restrictions:

No pre-built authentication boilerplate or starter kits

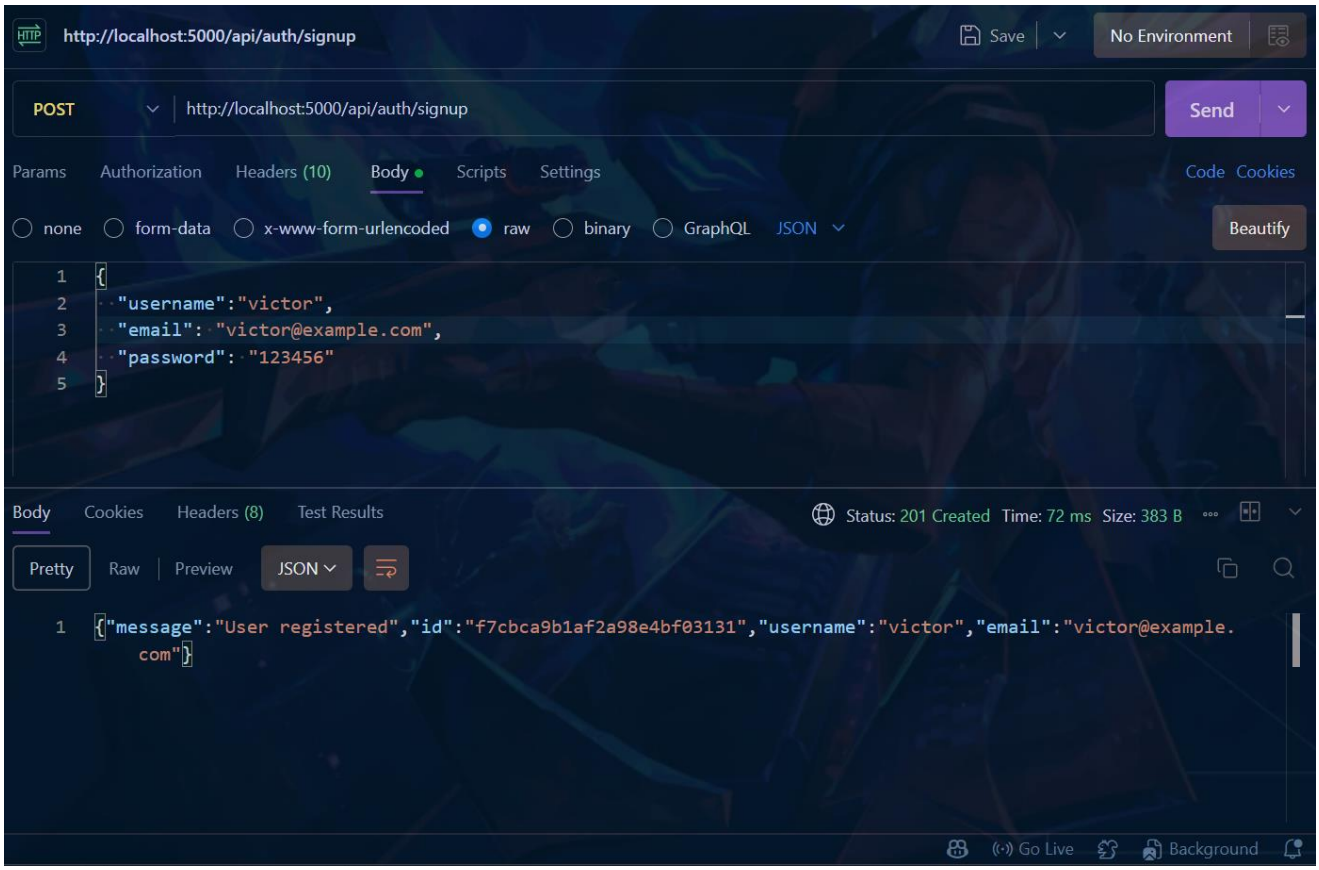
Only Node.js, Express, and the libraries listed above are allowed

Functional Requirements

- **Signup and Login System**
- **JWT Authentication**
- **Protected Routes**
- **Secure Notes API**
- **Real-Time Chat Functionality**
- **Event-Driven Features**

Expected Output

Sample console / API output showing the expected behaviour once each module is working correctly.



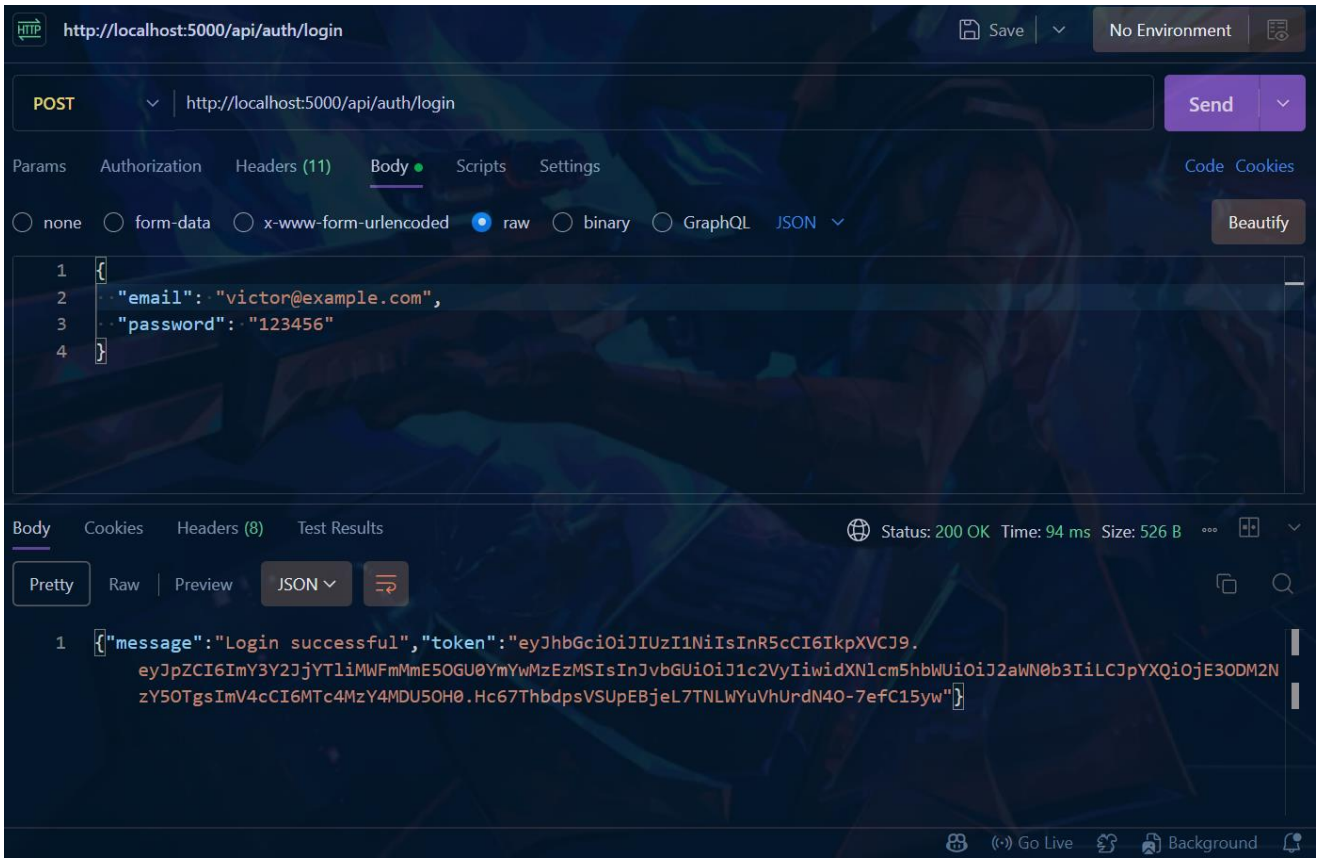


Fig 1. Signup and login responses (Postman / server console)

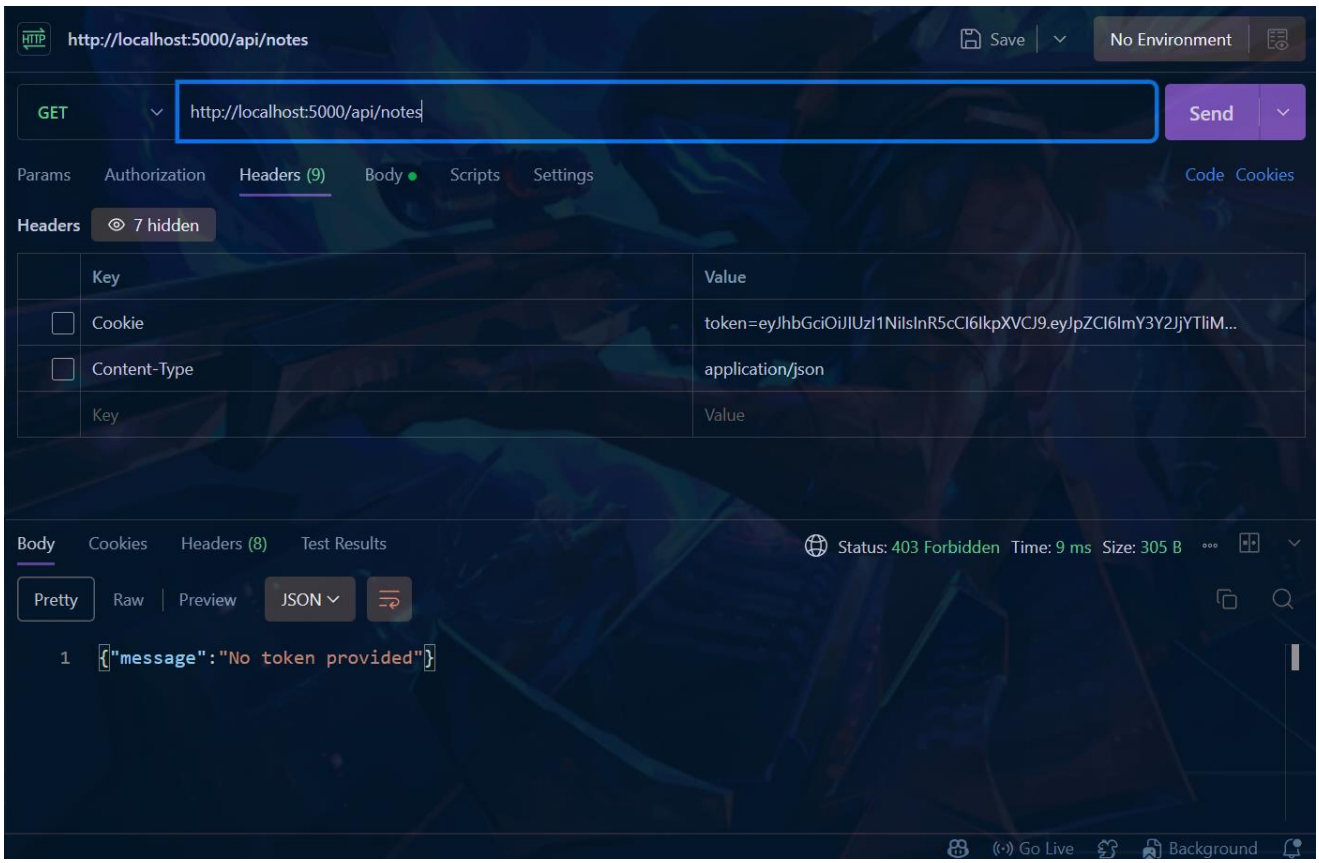




Fig 2. Notes API — blocked without a token, accessible with a valid JWT

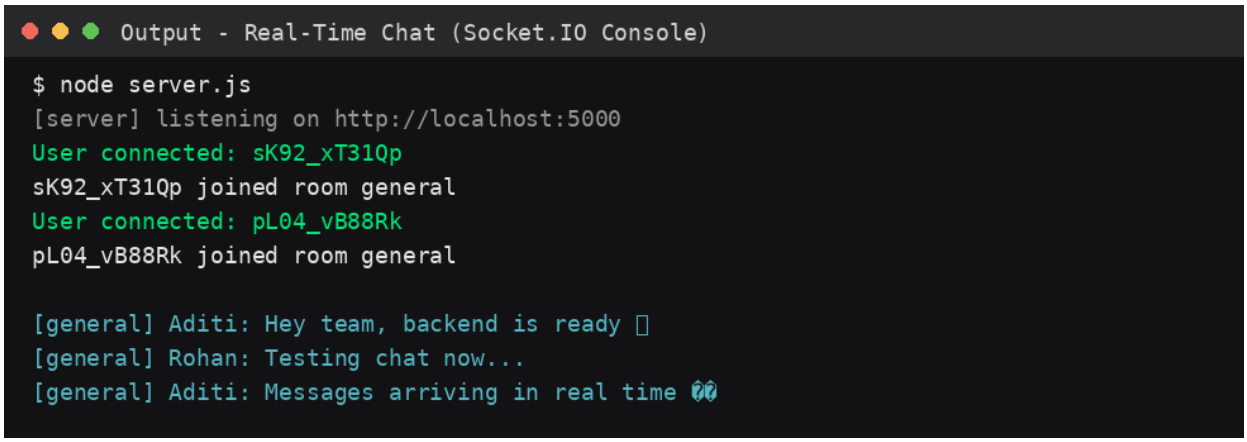


Fig 3. Real-time chat activity logged via Socket.IO

Concepts Covered

Authentication, Authorization, JWT, bcrypt, Protected Routes, RBAC, dotenv, WebSockets, Socket.IO, EventEmitter, Real-Time Communication

Rubric for Assignment – 4

Criteria	Marks
Functionality	1.5
API Design	0.5
Clean Code	0.5
Total	**2.5**

Happy Coding!